# What Do Vision Transformers Read? Executable Attention Reading Sets under Centre-Bias Control

**Author Name**
HKUST(GZ)
email@example.com

## Abstract

Attention maps are the most common way to look inside Vision Transformers (ViTs), but they are hard to verify and easily confused with the centre bias of photographs. We read, from unmodified pretrained ViTs, which tokens each image region attends to: following the fast-weight view of attention and the top-$k$ reading rule of MiTA, each of 16 regions keeps the 16 keys that its mean query scores highest. Because these reading sets can be executed, we can check that they are what the model uses: restricting all twelve layers to them keeps 72% of DeiT-S top-1 predictions, against 9% for random sets of the same size and 0.1% with no attention at all. Under tests that control for centre bias, three attention readouts of DeiT-S, DINO ViT-S and DeiT-B go to objects beyond the centre prior, follow objects moved off-centre and agree with human importance maps image by image; the models rely more on animal heads than bodies. Layer by layer, reading is most human-like in mid-to-late layers, and the last layer of the supervised ViTs loses most of this human-likeness. Tracing each query's attention budget in seven ViTs, we find that a few high-norm tokens take up to half of deep attention in supervised and top-$k$-trained models; neither top-$k$ training nor test-time registers moves the freed attention to objects, and these tokens do not explain the last-layer drift. As saliency maps, attention readouts remain less faithful than gradient-based GAE.

## 1 Introduction

Attention maps are cheap and look plausible, which makes them the default view into ViTs [Abnar and Zuidema, 2020, Caron et al., 2021]. Two problems limit what they tell us. A map is a picture, not a computation: nothing guarantees that the model relies on the highlighted tokens [Jain and Wallace, 2019, Wiegreffe and Pinter, 2019]. And objects sit near the centre of photographs, so a fixed centred Gaussian already correlates at 0.41 with ClickMe human importance maps [Fel et al., 2022], more than our probe on any model.

One attention layer computes the same function as a two-layer network whose first-layer weights are the keys, whose activation is the softmax and whose second-layer weights are the values [Schlag et al., 2021, Wen et al., 2026]. Each token is then a hidden neuron tied to an image position. MiTA [Wen et al., 2026] trains models that read only the top-$k$ such neurons per image region; we instead *read* these sets from unmodified pretrained ViTs and *execute* the model through them, so the reading is checked by the model's own output. Around it we build centre-controlled tests on three models and five datasets.

**Contributions.** (i) Executable reading sets that keep 62–83% of predictions when every layer reads only them, against 4–30% for random sets (Sec. 2). (ii) A centre-controlled protocol (Sec. 3). (iii) Findings that hold for three readouts: attention and the model's reliance go to objects and animal heads beyond the centre; reading is most human-like in mid-to-late layers and leaves the object in

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the last layer (Sec. 4.3). (iv) An attention-budget account: high-norm tokens absorb up to half of deep attention, and neither top-$k$ training nor test-time registers redirect it to objects (Sec. 4.4). (v) A negative result: attention readouts trail GAE on deletion faithfulness, and top-$k$ selection does not improve the map over dense attention (Sec. 4.2). All predictions and pass criteria were registered before the runs (Appendix D).

**Related work.** Rollout [Abnar and Zuidema, 2020] propagates attention through layers; GAE, AttnLRP, TokenTM and LibraGrad add gradients or relevance propagation [Chefer et al., 2021b,a, Achtibat et al., 2024, Wu et al., 2024b, Mehri et al., 2025]. We evaluate with deletion curves [Petsiuk et al., 2018] summarised by SRG [Blücher et al., 2024] under token removal [Jain et al., 2022, Mehri et al., 2025], SaCo [Wu et al., 2024a], the randomisation sanity check [Adebayo et al., 2018] and the Harmonization protocol [Fel et al., 2022]. Top-$k$ attention [Gupta et al., 2021, Wen et al., 2026] targets efficiency, and "attention received" scores tokens for pruning [Goyal et al., 2020, Kim et al., 2022, Wang et al., 2024]; our dense readout is this score, used here as an explanation. High-norm artefact tokens [Darcet et al., 2024] appear in supervised ViTs but not in the original DINO; they can be moved to a test-time register by editing a few FFN units [Jiang et al., 2025], and sinks can serve different computations [Fesser et al., 2026].

## 2 Method

**Reading sets.** For a layer and head with queries $q_i$ and keys $k_j$ ($j$ over 197 tokens), we pool the $14 \times 14$ patch grid into $m = 16$ regions ($4 \times 4$), take each region's mean query $\bar{q}_r$, and keep $S_r = \text{top-}\, k_j\, \bar{q}_r^\top k_j$ with $k = 16$. Every patch query reads only the set of its region; CLS reads its own top-$k$ set. To check them (*execution test*), we run the model with selected layers restricted to the sets and measure top-1 agreement with the unmodified model, against random sets of the same size.

**Three readouts.** For layers $\ell \in \mathcal{L}$, we score patch $j$ by the attention it receives:

$$s_j = \sum_{\ell \in \mathcal{L}} \sum_h \sum_i \frac{\exp(q_i^\top k_j / \sqrt{d})\, [j \in S_{r(i)}]}{\sum_{j' \in S_{r(i)}} \exp(q_i^\top k_{j'} / \sqrt{d})}. \tag{1}$$

The *probe* uses the reading sets as written; *dense attention received* drops the indicator (all 197 tokens, the token-pruning score [Goyal et al., 2020]); *CLS attention* keeps only the CLS query. Maps use layers 9–12 unless stated; none needs gradients or a class label.

## 3 Evaluation protocol

**Models and data.** DeiT-S/16 and DeiT-B/16 (supervised) [Touvron et al., 2021] and DINO ViT-S/16 with its official linear head [Caron et al., 2021]. ImageNet-S val (12,419 images with masks) [Gao et al., 2022], Hard ImageNet (750; small, off-centre objects) [Moayeri et al., 2022], Oxford Pet [Parkhi et al., 2012], ClickMe val (17,408 human importance maps) [Fel et al., 2022] and PartImageNet (24,080 part masks; images from the ImageNet *training* set) [He et al., 2022].

**Measures.** *Faithfulness*: patches are removed from every layer by dropping their tokens; SRG is the area between the least- and most-relevant-first deletion curves of top-1 agreement (0 for random scores). *Same-ring enrichment*: among patches at the same distance from the centre, the object share of a map's top patches minus the ring's object share; a centre prior scores 0. *Image-specific human alignment*: Spearman correlation with the image's own ClickMe map minus that with another image's map; a fixed prior scores 0. *Parts*: head enrichment is a map's head share over the head's area share; reliance is the log-probability drop when four patches of a part are removed. Intervals are 95% paired bootstrap intervals over images (1,000 resamples).

## 4 Results

### 4.1 The reading sets are what the model computes with

The test is the whole network: restricting all twelve layers keeps 72% (DeiT-S), 62% (DINO) and 83% (DeiT-B) at $k = 16$, against 9%, 4% and 30% for random sets and 0.1–0.2% with no attention at all. Single layers are no such test: one layer restricted to its sets keeps 92–96% of DeiT-S predictions

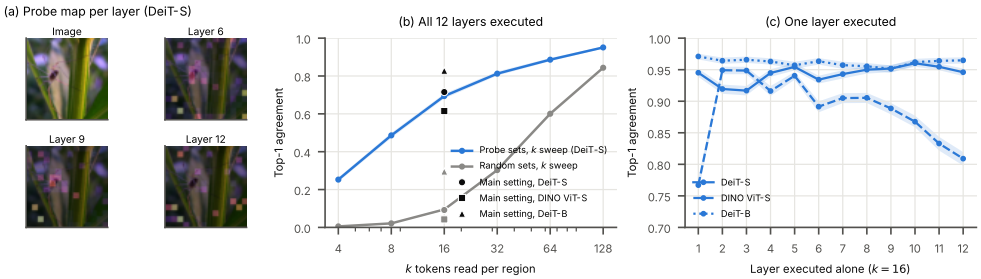



Figure 1: **Reading sets are executable.** (a) Single-layer probe maps of a random ImageNet-S image. (b) Top-1 agreement when all 12 layers read only the reading sets; the sweep uses a first-generation setting (CLS shares its region's set), markers the main setting at $k = 16$ for each model (grey: random sets). (c) One layer executed at a time (5,000 ImageNet images; bands: 95% intervals).

Table 1: **Attention readouts and gradient methods under the same tests** (S: DeiT-S, D: DINO ViT-S, B: DeiT-B). SRG and enrichment on ImageNet-S, human alignment on ClickMe, head enrichment on PartImageNet animals (1 = no preference). †Uses gradients and the predicted class. Bold: best attention readout.

| | SRG ↑ | | | Same-ring enrichment ↑ | | | ClickMe image-specific $\rho$ ↑ | | | Head enrichment | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | S | D | B | S | D | B | S | D | B | S | D | B |
| Probe (top-$k$ sets) | 0.201 | 0.235 | 0.150 | 0.114 | 0.158 | 0.067 | 0.135 | 0.199 | 0.092 | **1.54** | **1.82** | **1.30** |
| Dense attention received | **0.242** | 0.285 | **0.169** | **0.140** | 0.199 | **0.080** | **0.156** | 0.208 | **0.107** | 1.25 | 1.26 | 1.17 |
| CLS attention | 0.182 | **0.309** | 0.110 | 0.089 | **0.238** | 0.047 | 0.083 | **0.217** | 0.032 | 1.06 | 1.34 | 0.88 |
| Rollout | 0.146 | 0.172 | 0.030 | 0.110 | 0.144 | 0.045 | 0.141 | 0.171 | 0.050 | 1.09 | 1.04 | 1.01 |
| GAE† | 0.343 | 0.418 | 0.308 | 0.204 | 0.221 | 0.196 | 0.234 | 0.234 | 0.213 | 1.28 | 1.35 | 1.17 |
| Grad-CAM† | 0.341 | 0.316 | 0.274 | 0.186 | 0.120 | 0.170 | 0.208 | 0.145 | 0.172 | 1.10 | 1.18 | 1.04 |
| Integrated Gradients† | 0.037 | 0.230 | -0.001 | 0.008 | 0.152 | 0.004 | 0.008 | 0.195 | -0.010 | 1.01 | 1.23 | 0.96 |
| Centre Gaussian (reference) | 0.109 | 0.142 | 0.100 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 1.00 | 0.99 | 1.00 |

(Fig. 1c), but deleting that layer's attention outright already keeps 93–97% (69% at layer 1), so they only show which layers are harder to replace. Attention reading can therefore be treated as a measured quantity rather than a picture.

### 4.2 Attention readouts as saliency maps

Table 1 compares the readouts with GAE [Chefer et al., 2021a], Grad-CAM [Selvaraju et al., 2017] and Integrated Gradients [Sundararajan et al., 2017]. (1) A model-free centre Gaussian already reaches SRG 0.10–0.14 and beats Rollout on DeiT-B, so centre bias matters for faithfulness metrics too. (2) Among attention readouts, dense attention received is the most faithful on DeiT-S and DeiT-B and CLS attention on DINO, whose classifier reads the CLS token of the last four blocks; even the best of them trails GAE by 0.10–0.14 SRG. The same ordering holds on Hard ImageNet, Pet and an ImageNet subset (Appendix B). (3) The top-$k$ probe and dense attention are nearly the same map (median per-image Spearman 0.90). Dense attention is equal or better on faithfulness (higher in all 10 dataset–model pairs, by 0.006–0.050), object focus and human alignment; the probe's stronger head preference (1.54 vs. 1.25 on DeiT-S) does not make it better at naming the part the model relies on most (46.7% vs. 46.4%). The probe, CLS attention and Rollout pass the randomisation sanity check; Integrated Gradients fails it on all three models. We therefore use dense attention as the map and the top-$k$ sets as its verification.

### 4.3 What the models read

**Beyond the centre.** All readouts have positive same-ring enrichment (Table 1): reading favours objects even among patches equally far from the centre. Shifting an object 48 pixels towards a corner lowers the probe's object mass less than a centre prior's (by 0.021, 0.053, 0.010 for DeiT-S, DINO, DeiT-B), and 63–79% of the effect of removing object tokens survives matching the removed tokens' distance from the centre.

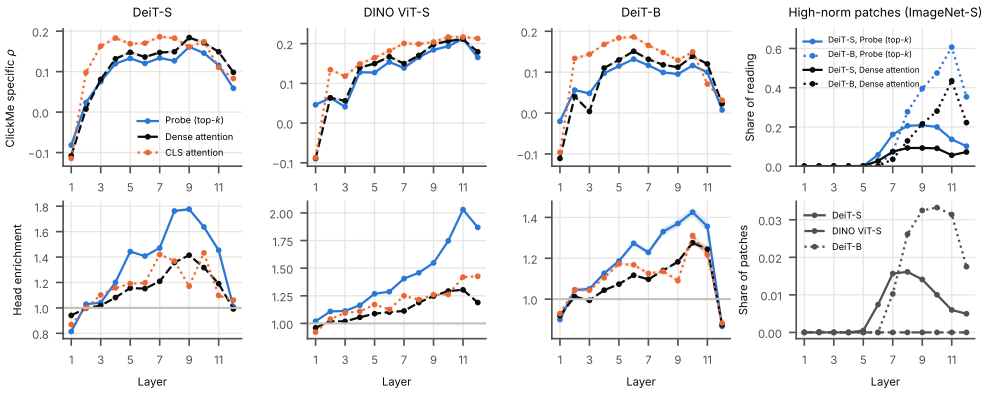



Figure 2: **Reading layer by layer.** Columns 1–3: image-specific human alignment (ClickMe, top) and head enrichment (PartImageNet, bottom) of single-layer maps for three readouts. Column 4: share of single-layer reading on high-norm patches (block-input norm $> 3\times$ the image's median) and share of such patches on ImageNet-S; DINO has none.

**Heads.** Removing four head patches lowers the log-probability of the predicted class more than removing four body patches of the same image ($+0.017$ [0.014, 0.020] DeiT-S, $+0.021$ DINO, $+0.008$ DeiT-B), 2–3.4 times as much for quadrupeds and bipeds; the probe, dense attention and GAE favour heads on all three models (enrichment $> 1$).

**Layers.** Human alignment and head preference rise through the network and peak in mid-to-late layers (DeiT-S layer 9, DINO layer 11, DeiT-B layers 6–10; Fig. 2). In the supervised models the last layer drops sharply (dense attention, layer 12 vs. the best of layers 8–11: ClickMe $-0.086$ DeiT-S, $-0.116$ DeiT-B; head enrichment $-0.42$, $-0.40$), whereas DINO drops little ($-0.032$; CLS attention $-0.004$). All three readouts show it (layer-curve correlation with the probe 0.77–0.99).

### 4.4 Where the attention budget goes

Each query spends a budget of 1 over the tokens. A registered follow-up (Appendix C) traced it on 11,919 ImageNet-S images in seven models with the same preprocessing, adding DeiT-T, DeiT-III-S [Touvron et al., 2022] and the MiTA-trained DeiT-S/T [Wen et al., 2026]. **High-norm tokens** take 30–50% of the budget in layers 9–12 of DeiT-B and both MiTA models (DeiT-B layer 11: 43% on 3% of patches); DeiT-T, DeiT-III-S and DINO have none. Their values are short (0.2–0.3 of an ordinary patch's). In DeiT-B the budget on them is used—blocking them costs 0.052 nats, more than blocking as many of the most-attended ordinary patches (0.037)—yet a fixed mean value serves as well as their own. **Two levers.** Top-$k$ training does not focus maps on objects as usually drawn: same-ring enrichment is lower for MiTA than DeiT at both sizes ($-0.039$, $-0.020$); excluding high-norm patches it is $-0.011$ at Small but $+0.037$ at Tiny. Moving DeiT-S's eight high-norm-making FFN units to a test-time register [Jiang et al., 2025] removes 92% of high-norm patches at unchanged accuracy, but the freed budget goes to the register (layer 9: $+5.9$ points), not the object ($+0.5$); enrichment rises by only 0.006, as if registers relocate sinks [Fesser et al., 2026]. In DeiT-B, 32 units remove only 45%. **The last layer.** Excluding high-norm patches, layer-12 enrichment is still below the best of layers 8–11 in all seven models (DINO $-0.057$), also with the register: high-norm tokens do not explain the drift.

## 5 Discussion and limitations

Executable reading sets make attention checkable, and centre-controlled tests show that ViTs read beyond where photographers put objects; a cleaner-looking map, from sparsity or registers, is not by itself evidence of more object focus. *Limitations.* Readouts are class-agnostic; part images are ImageNet training data; a layer-12 token keeps its patch only 15–25% of the time; each model was trained once, so intervals reflect images only. *Broader impact.* Plausible maps can create unjustified trust; we report centre references and negative results.

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## A Reproducibility

Pretrained weights are pinned by revision and checksum (timm DeiT-S/B, official DINO ViT-S/16 and its linear head). All runs use float32 without autocast and deterministic algorithms, on one NVIDIA A100 (main runs) or L4 (dense-attention and attention-budget runs; the probe re-run on L4 differs by at most 0.001 SRG). Each run stores its configuration, log, git commit and result tables; tables and figures are generated from those files by scripts (`scripts/paper_assets.py`). Faithfulness uses all ImageNet-S and Hard ImageNet images for DeiT-S and DINO and a fixed random 2,000-image subset (seed 0) for DeiT-B, Pet and the ImageNet subset. Removing a patch drops its token after the positional embedding, which equals masking it in every layer's attention (unit-tested to $2 \times 10^{-5}$ in the logits).

Table 2: SRG on every dataset and model (S: DeiT-S, D: DINO, B: DeiT-B).

| Method | ImageNet-S S | ImageNet-S D | ImageNet-S B | Hard ImageNet S | Hard ImageNet D | Hard ImageNet B | Pet S | Pet D | ImageNet S | ImageNet D |
|---|---|---|---|---|---|---|---|---|---|---|
| Probe (top-$k$ sets) | 0.201 | 0.235 | 0.150 | 0.177 | 0.192 | 0.136 | 0.252 | 0.328 | 0.205 | 0.235 |
| Dense attention received | 0.242 | 0.285 | 0.169 | 0.198 | 0.233 | 0.142 | 0.278 | 0.375 | 0.241 | 0.277 |
| CLS attention | 0.182 | 0.309 | 0.110 | 0.135 | 0.242 | 0.100 | 0.179 | 0.387 | 0.179 | 0.301 |
| Rollout | 0.146 | 0.172 | 0.030 | 0.134 | 0.126 | 0.025 | 0.210 | 0.271 | 0.149 | 0.170 |
| GAE$^\dagger$ | 0.343 | 0.418 | 0.308 | 0.325 | 0.397 | 0.316 | 0.416 | 0.480 | 0.361 | 0.425 |
| Grad-CAM$^\dagger$ | 0.341 | 0.316 | 0.274 | 0.315 | 0.328 | 0.279 | 0.360 | 0.351 | 0.345 | 0.335 |
| Integrated Gradients$^\dagger$ | 0.037 | 0.230 | -0.001 | 0.034 | 0.183 | 0.012 | 0.060 | 0.298 | 0.047 | 0.221 |
| Centre Gaussian (reference) | 0.109 | 0.142 | 0.100 | 0.064 | 0.115 | 0.061 | 0.160 | 0.202 | 0.114 | 0.143 |
| Random (reference) | -0.001 | -0.001 | -0.003 | 0.002 | -0.007 | -0.003 | 0.010 | 0.001 | 0.001 | -0.000 |

# B Additional results

# C Attention-budget study

Hypotheses, a minimum relevant difference of 0.03 in same-ring enrichment, controls and gates were registered before any new run; judgements use paired bootstrap intervals over images at 99.5% (Bonferroni over 10 comparisons). All models use Resize 256 and a 224 centre crop (accuracy changes $\leq 0.18$ points; all seven reproduce their published accuracy, MiTA-DeiT-T 71.14%). For paired models, the union of both models' high-norm positions in layers 9–12 is excluded. H1 (DINO more object-focused than all six supervised models) failed: $+0.036$ to $+0.065$ for five, but $-0.002$ [$-0.005$, 0.000] against MiTA-DeiT-T. H2 (top-$k$ training gives no relevant gain) held at Small ($-0.011$ [$-0.013$, $-0.008$]) and failed at Tiny ($+0.037$ [0.035, 0.040]); another supervised recipe (DeiT-III-S − DeiT-S) differs by only $-0.002$. H6 (moving the register units gives no relevant gain) held for DeiT-S ($+0.006$ [0.005, 0.007]); DeiT-B failed the registered gate (units must remove $\geq 80\%$ of high-norm patches) and was not tested. Sink type (no-op, constant bias or broadcast) stayed undecided under the registered thresholds for both models.

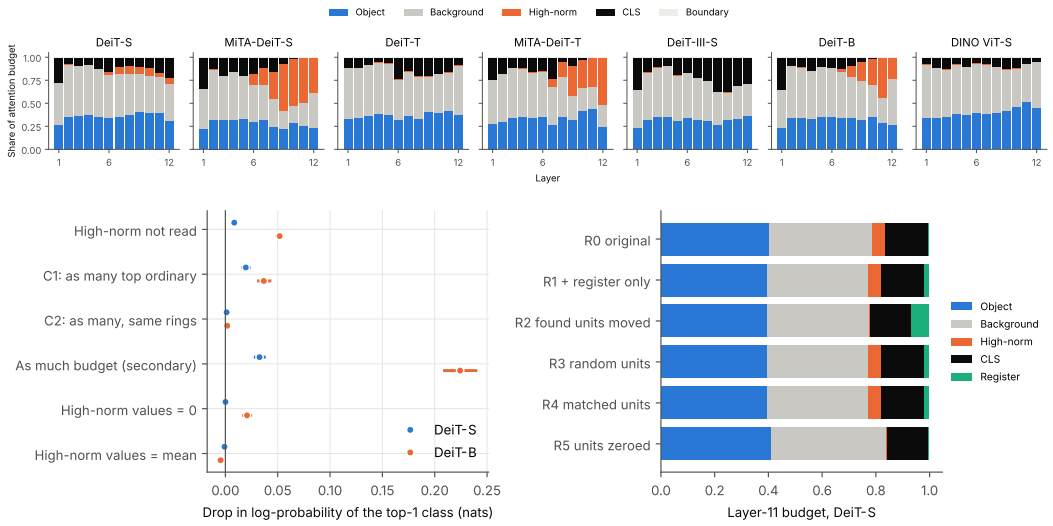



Figure 3: Top: share of each layer's attention budget (all queries) on object, background, high-norm patches and CLS. Bottom left: drop in log-probability when no query reads the high-norm patches, as many top ordinary or ring-matched patches, or when their values are zeroed or replaced by a mean. Bottom right: DeiT-S layer-11 budget under test-time register conditions (R2: the eight found units moved to the register; R3/R4: random or activation-matched units).

# D Registered predictions

Every prediction and its pass criterion was written down before the runs. Of the 32 directional predictions in our ledger, 20 held, 7 held partly and 5 failed; H1, H2 and H6 of the attention-budget study are described in Appendix C. Failed: the probe is *not* as faithful as GAE (all 10 pairs); the

centre prior's head enrichment is about 1, not below 1; and, in a direction favourable to us, the DeiT-S and DeiT-B probes do show object preference beyond the centre (ring enrichment $+0.114$, $+0.067$) where we expected none. Partly held: the probe beats CLS attention on DeiT-S and DeiT-B but not on DINO; Integrated Gradients fails the sanity check and does not beat random scores on DeiT-B; "dense attention is comparable to the probe" held for SRG but not fully for object focus and human alignment, where dense attention is better; the high-norm share rises at the last layer for DeiT-B but not DeiT-S.